Physical Chemistry CUME, March 3, 2017

Chemical Equilibrium

Consider the following reaction:

$$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$$

1a. Write an expression for the "thermodynamic equilibrium constant" (K) in terms of the activities of N_2, H_2, and NH_3 (i.e., a_{N_2}, a_{H_2}, and a_{NH_3}). How do the "reaction quotient" (Q) and K differ in their definitions? [10 points]

1b. Write an expression for "reaction Gibbs energy" ($\Delta_r G$) in terms of Q and K. Hint: Start with writing an expression for $\Delta_r G$ in terms of the "standard reaction Gibbs energy" ($\Delta_r G^{\ominus}$) and Q. [10 points]

1c. Assuming the "activity coefficients" are close to unity, write an expression for the equilibrium constant (K) in terms of the equilibrium concentrations of N_2, H_2, and NH_3? Hint: K is unitless. [10 points]

**Assuming the overall pressure is low enough for the gases to be treated as ideal, answer the following questions using the data in Table 1:*

2. Find $\Delta_r G$ at 298 K, when the partial pressures of N_2, H_2, and NH_3 are 3.0, 1.0, and 4.0 *bar*, respectively? What is the spontaneous direction of the reaction in this case? [20 points]

3. Find the equilibrium constant at any arbitrary temperature. Hint: Start with finding K_1, the equilibrium constant at $T_1 = 298\ K$. A relation between K_1 and K_2, the equilibrium constant at any arbitrary temperature T_2, can be derived from the van 't Hoff equation: $\frac{d \ln K}{d(\frac{1}{T})} = -\frac{\Delta_r H^{\ominus}}{R}$. [30 points]

4. Predict the effect of an n-fold increase in pressure on the equilibrium of the above reaction in terms of the equilibrium mole fractions of reactants and product. [20 points]

Table 1. Thermodynamic data (all values are for 298 K)

	$M/(g\ mol^{-1})$	$\Delta_f H^{\ominus}(kJ\ mol^{-1})$	$\Delta_f G^{\ominus}(kJ\ mol^{-1})$	$S_m^{\ominus}(kJ\ mol^{-1})$
$H_2(g)$	2.02	0	0	130.7
$N_2(g)$	28.01	0	0	191.6
$NH_3(g)$	17.03	−46.1	−16.5	192.5